IT technician Windows tools and ISO files

An IT technician typically carries a well-equipped USB flash drive and external hard disk with essential Windows tools and ISO files for troubleshooting, maintenance, installation, and recovery. Here's a comprehensive list:

A. USB Flash Drive (Bootable Tools & Portables - 16GB to 64GB or more)

1. Bootable Utilities (Multiboot or individual tools)

Ventoy / Rufus – Tool to create multiboot USBs.

Hiren's BootCD PE – All-in-one rescue utility (partitioning, backup, recovery, etc.).

Sergei Strelec's WinPE – Powerful WinPE environment with tools for maintenance and repair.

Gandalf's WinPE – Similar to Hiren's with extra utilities.

MemTest86 – RAM diagnostics.

DBAN (Darik's Boot and Nuke) – Secure data erasure tool.

2. Portable Software (No installation needed)

CPU-Z / HWMonitor – Hardware diagnostics.

CrystalDiskInfo – Disk health checker.

Malwarebytes / AdwCleaner (Portable) – Malware cleanup.

Autoruns / Process Explorer – Deep system inspection tools from Sysinternals.

DriverPack Solution (Lite) – For offline driver installation.

ProduKey / NirSoft Utilities – Keyfinder and system info tools.

B. External Hard Disk (Larger ISOs, Backups, Repositories - 500GB to 2TB+)

1. Operating System ISO Files

Windows ISO files:

Windows 11 (latest version)

Windows 10 (Home, Pro, Enterprise)

Windows 7 (for legacy support)

Windows Server (2016, 2019, 2022)

Linux Distros (for rescue/dual boot):

Ubuntu, Kali Linux, GParted Live

2. Recovery & Repair ISOs

Macrium Reflect Rescue ISO – For system image backups and restores.

Acronis True Image ISO – Backup and restore solutions.

Windows Recovery ISO / Windows ADK – For recovery environment.

3. Drivers & Software Repositories

Offline driver packs – e.g., Snappy Driver Installer.

Essential Windows software installers – Chrome, VLC, WinRAR, .NET, Java, etc.

System image backups – Custom OS images for rapid deployment.

4. Windows Deployment & Management Tools

Microsoft Deployment Toolkit (MDT)

Windows Assessment and Deployment Kit (ADK)

Sysprep-prepared Windows images

Unattended installation scripts

Optional But Useful

Password reset tools (Offline NT Password & Registry Editor, PCUnlocker)

Network tools – Wireshark, Nmap, Angry IP Scanner.

Cloning tools – Clonezilla, AOMEI Backupper.

Partition tools – MiniTool Partition Wizard, EaseUS Partition Master.

List of Windows tools and ISO files and their uses

	Category	Tool/ISO Name	Uses
1	Bootable Utilities	Ventoy / Rufus	Create multiboot USB Windows tools
2	Bootable Utilities	Hiren's BootCD PE	System repair, partitioning, password reset
3	Bootable Utilities	Sergei Strelec's WinPE	Advanced system rescue, includes backup tools
4	Bootable Utilities	Gandalf's WinPE	Alternative rescue disk with similar tools to Hiren's
5	Bootable Utilities	MemTest86	RAM diagnostics for troubleshooting hardware issues
6	Bootable Utilities	DBAN	Securely wipe student/staff data before reinstall
7	Portable Software	CPU-Z / HWMonitor	Check hardware specs and thermal status of school PCs
8	Portable Software	CrystalDiskInfo	Check health of hard disks in lab and office PCs
9	Portable Software	Malwarebytes / AdwCleaner	Remove malware and adware from infected systems
10	Portable Software	Autoruns / Process Explorer	Analyze startup items and running processes
11	Portable Software	DriverPack Solution (Lite)	Install drivers offline after Windows installation
12	Portable Software	ProduKey / NirSoft Utilities	Recover product keys from existing installations
13	Portable Software	Windows 11 ISO	Install latest Windows version for staff or lab PCs
14	Portable Software	Windows 10 ISO	Reinstall common version used in schools
15	Operating System ISOs	Windows 7 ISO	Legacy OS support for older systems
16	Operating System ISOs	Windows Server ISO	Install or repair Windows Server (for shared resources)
17	Operating System ISOs	Ubuntu / Kali Linux ISO	Linux OS for training or dual-boot with Windows
18	Operating System ISOs	GParted Live ISO	Resize/repair partitions in lab machines
19	Operating System ISOs	Macrium Reflect Rescue ISO	Restore system backups on school computers
20	Recovery & Repair ISOs	Acronis True Image ISO	Backup/restore complete images of staff systems
21	Recovery & Repair ISOs	Windows Recovery ISO	Use for repairing broken Windows installations
22	Recovery & Repair ISOs	Windows ADK ISO	Create custom recovery environment
23	Recovery & Repair ISOs	Snappy Driver Installer	Offline drivers for PCs with no internet access
24	Drivers & Software Repositories	Essential Software Installers	Install common tools (browsers, PDF readers, etc.)
25	Drivers & Software Repositories	System Image Backups	Quick recovery of full school PC images
26	Drivers & Software Repositories	Microsoft Deployment Toolkit (	Automate Windows setup for multiple PCs
27	Windows Deployment Tools	Windows ADK	Required for image customization and deployment

28	Windows Deployment Tools	Sysprep-prepared Windows Images	Restore custom school OS image
29	Windows Deployment Tools	Unattended Installation Scripts	Automate installation with school-specific settings
30	Windows Deployment Tools	Offline NT Password Editor	Reset forgotten staff/admin passwords
31	Optional Tools	PCUnlocker	Unlock PCs for students or teachers
32	Optional Tools	Wireshark / Nmap / Angry IP Scanner	Diagnose network issues in computer labs
33	Optional Tools	Clonezilla / AOMEI Backupper	Clone systems or backup staff computers
34	Optional Tools	Partition Tools	Resize partitions for dual boot or repairs

How to create and use a Hiren's BootCD PE USB to reset windows password or create a new guest user.

□ What You'll Need:

1. A **working computer** with internet access.
2. A **USB flash drive** (at least 2GB).
3. A tool called **Rufus** (to make the USB bootable).
4. **Hiren's BootCD PE ISO** file.

▼ Step 1: Download Tools

□ Hiren's BootCD PE ISO:

- Go to https://www.hirensbootcd.org/download/
- Download the **Hiren's BootCD PE x64 ISO**.

□ Rufus:

- Visit https://rufus.ie
- Download and run **Rufus** (portable version is fine).

💽 Step 2: Create Bootable USB

1. **Insert your USB drive** (⚠ all data on it will be erased).
2. Open **Rufus**.
3. Under "**Device**," select your USB drive.
4. Under "**Boot selection**," choose "Disk or ISO image" and click **SELECT**.
5. Choose the Hiren's BootCD PE ISO you downloaded.
6. Click **Start** and let it create the bootable USB.

🔄 Step 3: Boot from USB on Locked PC

1. Insert the USB into the **locked computer**.
2. Power it on and press the **boot menu key** (usually **F12**, **Esc**, or **F9** depending on your PC brand).
3. Select the USB drive to boot from.

🛠 Step 4: Use Built-In Tool to Reset or Create User

Once booted into Hiren's BootCD PE:

1. You'll see a desktop environment similar to Windows.
2. Click on **Start > Utilities > Security > Lazesoft Password Recovery**.
3. Launch **Lazesoft Password Recovery**:
   - Select **Reset Windows Password**.
   - Choose the Windows installation (usually C:\Windows).
   - It will list all user accounts.
4. You can:
   - **Reset the admin password** (recommended).
   - Or **create a new user** and make it a member of the **Guests** group.

Example: Create a user named `GuestUser` and assign it to **Guests group**.

✅ After You're Done:

1. **Remove the USB**.
2. **Restart the computer** and boot normally.
3. Log in with your new or reset account.

You can safely remove the USB drive after the password reset is completed—**but** only after **you've closed the password reset tool and shut down or restarted the computer properly.**

Here's the right sequence:

▯ After Resetting the Password or Creating a New User:

1. **Close Lazesoft Password Recovery** or any other tool you used.
2. Click **Start > Power > Restart** or **Shut Down** in the Hiren's BootCD environment.
3. As soon as the screen goes black during the shutdown or restart process, **remove the USB drive**.
4. The system will then boot into **your regular Windows installation** (not from USB).

▯ Important Tips:

- **Do not** just yank the USB out while the system is still running from it—it might be in use and could cause errors.
- If your computer boots back into Hiren's again, it means the USB is still set as the first boot option. Just remove the USB and reboot.

You can use the same Hiren's BootCD PE USB to reset a Windows 7 password, but with a small caveat:

▯ What Works on Windows 7

Hiren's BootCD PE (based on Windows 10) **does support password reset for Windows 7, 8, 10, and 11**, using tools like:

- **Lazesoft Password Recovery**
- **Offline NT Password & Registry Editor**
- **NTPWEdit**

These tools work by **accessing the SAM (Security Account Manager)** file on the Windows drive, which exists in all Windows versions.

▯ How to Reset Windows 7 Password with It:

1. **Boot the locked Windows 7 computer from the Hiren's USB**.
2. Open a password tool such as:
   - **Lazesoft Password Recovery**
   - Or **NTPWEdit** (found under `Start > Utilities > Security`)
3. Select the correct **Windows 7 installation drive** (usually `C:\Windows`).
4. Select the user whose password you forgot.
5. Click **"Reset/Unlock"** or **"Clear Password"**, then save and exit.

📝 Notes:

- After clearing the password, you can log in to the user account by **leaving the password field empty**.
- Windows 7 **Guest accounts** are even more limited than on Windows 10, so you may prefer creating a standard user instead.

How to create new guest user on windows 10 when the screen password is forgotten?

Use Safe Mode with Command Prompt (if accessible)

This method allows you to create a new user via Command Prompt **if** you can boot into Safe Mode with Command Prompt.

Steps:

1. **Power on the computer**, and as soon as it starts to load Windows, press and hold the **power button** to shut it down.
2. Do this **2-3 times** until **"Preparing Automatic Repair"** appears.
3. Select **Advanced Options > Troubleshoot > Advanced Options > Startup Settings > Restart**.
4. Press **F6** for **Safe Mode with Command Prompt**.

In Command Prompt, type the following commands:

```
net user GuestUser P@ssword123 /add
net localgroup users GuestUser /add
```

This creates a basic user. To give guest-like permissions:

```
net localgroup guests GuestUser /add
```

⊘ Do **not** add this user to "Administrators" if you want it to remain a limited account.

Ventoy USB, Windows Repair tool

Recommended iso files

Use Ventoy for ISO-based tools like:

- Hiren's BootCD PE
- Sergei Strelec
- Gandalf's PE
- SystemRescue
- UBCD (Ultimate Boot CD)
- AIO (All-In-One)
- MediCat USB
- SystemRescue
- Rescatux

Use AIO Boot separately if you want an advanced multiboot system with built-in installers and more complex setup.

🛠 Top Multiboot Utilities for Ventoy

Here's a curated list of powerful multiboot utilities you can add to your Ventoy USB drive. These tools are invaluable for system recovery, diagnostics, partitioning, and more. Each entry includes a brief description and a direct download link for the ISO file.

1. Hiren's BootCD PE (Windows PE)

Purpose: A comprehensive Windows-based recovery environment packed with tools for data recovery, partitioning, antivirus scanning, and system diagnostics.

Download: Hiren's BootCD PE ISO

2. Sergei Strelec WinPE

Purpose: A Windows PE-based bootable environment featuring a wide array of utilities for backup, recovery, partitioning, and diagnostics.

Download: Sergei Strelec's WinPE ISO

3. Gandalf's Windows PE

Purpose: A Windows PE bootable environment similar to Hiren's, offering a suite of tools for system maintenance and recovery.

Download: Gandalf's Windows 10 PE ISO

4. MediCat USB

Purpose: A multiboot USB tool that includes a variety of utilities for system repair, diagnostics, and malware removal.

5. SystemRescue

Purpose: A Linux-based system rescue toolkit designed for repairing unbootable or damaged systems, with tools for disk partitioning, data recovery, and more.

Download: SystemRescue ISO

6. Ultimate Boot CD (UBCD)

Purpose: A bootable CD that consolidates numerous diagnostic tools for hardware testing, memory checks, and system analysis.

Download: Ultimate Boot CD ISO

7. AIO Boot

Purpose: A versatile multiboot tool that supports booting various operating systems and tools, including Windows and Linux distributions.

Download: AIO Boot ISO

8. Rescatux

Purpose: A Debian-based live distribution featuring a graphical wizard for rescuing broken GNU/Linux and Windows installations and boot loaders.

Download: Rescatux ISO

📁 Organizing ISOs on Your Ventoy USB Drive

To keep your Ventoy USB organized, you can create folders for different categories:

```
Ventoy USB Drive (E:)\
├── Windows\
│   ├── Windows10.iso
│   └── Windows11.iso
├── Recovery_Tools\
│   ├── HirenPE.iso
│   ├── Strelec.iso
│   └── GandalfPE.iso
├── Linux_Tools\
│   ├── SystemRescue.iso
│   └── Rescatux.iso
├── Diagnostics\
│   └── UBCD.iso
└── Multiboot_Tools\
    └── AIOBoot.iso
```

Ventoy will automatically detect and list all ISO files, even if they're organized into folders.

Here are direct download links for various ISO tools you can add to your Ventoy USB drive, organized by category:

🛠 Recovery & Windows PE Tools

1. Hiren's BootCD PE (Windows 11 PE x64)

Download: hirensbootcd.org/download, Size: ~3.3 GB

Features: Includes tools like Recuva, Lazesoft Recovery, and CrystalDiskInfo.

2. Gandalf's Windows 10 PE x64 (Redstone 7)

Download: archive.org/details/gandalfs-win-10-pe-x64-redstone-7-20h1, Size: ~4.2 GB

Features: A comprehensive Windows PE environment with various utilities.

3. Gandalf's Windows 11 PE x64 (RS8 22H2)

Download: archive.org/details/gandalfs-win-11-pex-64-rs-8-22-h-2-summer-2023, Size: ~11.2 GB

Features: Updated tools and drivers for modern hardware.

□ Antivirus Rescue Disks

4. Kaspersky Rescue Disk 18

Download: kaspersky.com/downloads/free-rescue-disk, Size: ~600 MB

Features: Bootable antivirus scanner with GUI and registry editor.

5. Bitdefender Rescue CD (Discontinued)

Download: archiveos.org/bitdefender, Size: ~872 MB

Note: No longer updated; consider alternatives like ESET SysRescue Live.

🐧 Linux-Based Rescue & Partition Tools

6. SystemRescue (formerly SystemRescueCd)

Download: system-rescue.org/Download, Size: ~700 MB

Features: Includes GParted, fsarchiver, and other system utilities.

7. GParted Live

Download: gparted.org/download.php, Size: ~590 MB

Features: Graphical partition editor for managing disk partitions.

8. Rescatux

Download: sourceforge.net/projects/rescatux/files/rescatux-0.74/rescatux-0.74.iso/download

Size: ~700 MB

Features: GUI-based tool for repairing GRUB and Windows bootloaders.

□ Operating System ISOs

9. Ubuntu Desktop

Download: ubuntu.com/download/desktop, Size: ~5.8 GB

Features: Full-featured Linux desktop environment.

10. Ubuntu Server

Download: ubuntu.com/download/server, Size: ~1.5 GB

Features: Optimized for server deployments.

📁 Organizing ISOs on Ventoy USB

To keep your Ventoy USB organized, create folders like:

```
Ventoy_USB/
├── Windows/
│   ├── Windows10.iso
│   └── Windows11.iso
├── Recovery/
│   ├── Hiren's_BootCD_PE.iso
│   ├── Gandalf_Win10_PE.iso
│   └── Gandalf_Win11_PE.iso
├── Antivirus/
│   ├── Kaspersky_Rescue.iso
│   └── Bitdefender_Rescue.iso
├── Linux_Tools/
│   ├── SystemRescue.iso
│   ├── GParted_Live.iso
│   └── Rescatux.iso
├── OS/
│   ├── Ubuntu_Desktop.iso
│   └── Ubuntu_Server.iso
│
```

Note: Ventoy will detect ISOs in subfolders, so organizing them won't affect functionality.

Antiviruses

Kaspersky Rescue Disk

To start rescuing your PC, follow these easy steps:

I. Save the Kaspersky Rescue Disk software to a USB memory device or a DVD (to learn how to do this, see https://support.kaspersky.com/help/krd/2024/en-us/272239.htm).
II. Boot up your PC – from the storage device that contains Kaspersky Rescue Disk.
III. Update the antivirus databases.
IV. Run a system scan on your PC and then follow the instructions on your screen.

Review

Here's a list of ISO files that can be used to repair Windows 10 boot problems using Ventoy on a USB flash drive:

1. Windows 10 Installation ISO (Official Microsoft)

- Purpose: Repair startup, restore system images, access Recovery Environment (WinRE).

- How to get it:

  - Download from [Microsoft's official site](https://www.microsoft.com/software-download/windows10) (Media Creation Tool).

  - Use Rufus or Media Creation Tool to create an ISO.

2. Windows 10 Recovery Disk (WinRE)

- Purpose: Access advanced repair options (Startup Repair, Command Prompt, System Restore).

- How to get it:

  - Create via Control Panel > Recovery > Create a recovery drive (requires a working Windows PC).

  - Extract boot.wim and convert to ISO using tools like Oscdimg.

3. Hiren's BootCD PE

- Purpose: Includes Mini Windows 10 PE with tools like:
  - Bootrec.exe (fix MBR/BCD).
  - DISM (repair Windows image).
  - TestDisk (fix partitions).
- Download: [Hiren's BootCD PE](https://www.hirensbootcd.org/)

4. Medicat USB (Portable Windows PE)

- Purpose: Contains multiple boot repair tools:
  - Automatic Repair (like Windows RE).
  - NTBootFix (fix bootloader).
  - Partition Wizard (fix disk errors).
- Download: [Medicat USB](https://medicatusb.com/)

5. Ultimate Boot CD (UBCD)

- Purpose: Includes Parted Magic, TestDisk, and Boot Repair Utilities.
- Download: [UBCD](https://www.ultimatebootcd.com/)

6. SystemRescue (Linux-based but useful for disk repairs)

- Purpose: Fix NTFS, recover partitions, repair GRUB (if dual-booting).
- Download: [SystemRescue](https://www.system-rescue.org/)

7. Boot Repair Disk (For BCD/MBR issues)

- Purpose: Automatically fixes bootloader issues (like bootrec /fixmbr).
- Download: [Boot Repair Disk](https://sourceforge.net/projects/boot-repair-cd/)

How to Use Ventoy for Boot Repair:

1. Download any of the above ISOs.

2. Copy them to your Ventoy USB drive.

3. Boot from the USB and select the ISO you need.

4. Follow repair steps (e.g., Startup Repair, bootrec /fixboot, chkdsk).

Best ISO for Common Problems:

- MBR/BCD Corruption → Windows 10 ISO (bootrec commands).

- Missing Bootloader → Hiren's BootCD PE or Boot Repair Disk.

- Corrupt System Files → Windows 10 ISO (DISM / SFC).

- Partition Issues → Medicat or UBCD.

Hiren's BootCD

Repair Windows Startup Problems Hiren's BootCD on Ventoy USB

Hiren's BootCD includes powerful tools to diagnose and fix Windows boot issues, including:

- Corrupted Bootloader
- Missing System Files
- Damaged Registry
- Failing Hard Drive

🔧 Step-by-Step Guide: Fixing Windows Startup with Hiren's BootCD

1□ Boot into Hiren's BootCD from Ventoy

1. Restart PC → Enter Boot Menu (usually F12, ESC, or DEL).
2. Select Ventoy USB → Choose Hiren's BootCD ISO.
3. Select "Mini Windows 10" (or "Boot from RAM" if available).

2□ Repair Windows Startup

Option A: Automatic Repair (Recommended)

- Open "Startup Repair" (in Hiren's menu) → Select your Windows partition.

- Let it scan and fix boot issues automatically.

Option B: Manual Boot Repair (Advanced)

1. Open Command Prompt (from Hiren's tools).
2. Run these commands (adjust C: if Windows is on a different drive):
   bootrec /fixmbr
   bootrec /fixboot
   bootrec /scanos
   bootrec /rebuildbcd
   - This rebuilds the bootloader if corrupted.

Option C: Fix Corrupted System Files

1. Run DISM (if Windows is partially bootable):

   dism /image:C:\ /cleanup-image /restorehealth

2. Run SFC to repair system files:

   sfc /scannow /offbootdir=C:\ /offwindir=C:\Windows

Option D: Restore Registry Backup

- Navigate to:

  C:\Windows\System32\config\RegBack

- Copy backup files (DEFAULT, SAM, SECURITY, SOFTWARE, SYSTEM) to:

  C:\Windows\System32\config

⬙ Other Useful Hiren's Tools for Startup Issues

- HDD Regenerator → Fix bad sectors preventing boot.

- TestDisk → Recover lost partitions.

- NTBOOTautofix → Automatic boot repair for Win 7/10/11.

- UltraBlue Screen Viewer → Analyze BSOD errors.

□ Troubleshooting Tips

□ "Bootrec /fixboot Access Denied"?

- Run:

bootsect /nt60 sys

□ BCD errors?

- **Rebuild BCD manually:**

bcdedit /export C:\BCD_Backup

ren C:\Boot\BCD BCD.old

bootrec /rebuildbcd

🎯 Final Verdict

□ Hiren's BootCD + Ventoy = Ultimate Repair Combo

- Fixes MBR/BCD corruption, missing system files, and registry errors.

- Works even if Windows won't boot at all.

🔧 Common Windows Problems & How to Fix Using Hiren's BootCD (Via Ventoy USB)

Here's a quick-reference guide for frequent Windows issues and the best Hiren's tools to solve them:

1□ Windows Won't Boot (Black Screen/Infinite Loop)

Possible Causes:

- Corrupt bootloader
- Dead hard drive
- Missing system files

Fix:

1. Boot Hiren's Mini Windows 10.
2. Run NTBOOTautofix (automatic boot repair).
3. OR use Command Prompt:

```
bootrec /fixmbr
bootrec /fixboot
bootrec /rebuildbcd
chkdsk C: /f /r
```

2□ Blue Screen of Death (BSOD) Errors

Common Errors:

- CRITICAL_PROCESS_DIED

- MEMORY_MANAGEMENT

- KERNEL_SECURITY_CHECK_FAILURE

Fix:

1. Use BlueScreenView (in Hiren's) to analyze crash dumps.

2. Test RAM with MemTest86 (if crashes persist).

3. Run **SFC/DISM** in CMD:

```
sfc /scannow /offbootdir=C:\ /offwindir=C:\Windows
dism /image:C:\ /cleanup-image /restorehealth
```

3 Hard Drive Not Detected/Corrupt

Symptoms:

- "Disk Unknown" in BIOS
- Slow performance, clicking noises

Fix:

1. Run HDD Regenerator (fix bad sectors).
2. Use TestDisk (recover lost partitions).
3. Check SMART status with CrystalDiskInfo.

4 Forgotten Windows Password

Solution:

1. Use NTPWEdit (in Hiren's) to reset passwords.
2. OR Offline NT Password & Registry Editor (for local accounts).

5 Virus/Malware Infection

Signs:

- Unusual slowdowns
- Pop-ups, ransomware

Fix:

1. Run Kaspersky Virus Removal Tool (in Hiren's).
2. Use Malwarebytes Chameleon (if malware blocks scans).

6 Accidentally Deleted Files

Recovery:

1. Use Recuva (simple undelete).

2. OR PhotoRec (for deep recovery of formatted drives).

7□ Slow Performance (Not Hardware-Related)

Culprits:
- Bloatware
- Fragmented HDD

Fix:
1. Run CCleaner (clean junk files).
2. Use Defraggler (optimize HDDs).

8□ Wi-Fi/Network Issues

Quick Fixes:

1. Reset TCP/IP stack in CMD:

netsh int ip reset

netsh winsock reset

2. Use WirelessKeyView (recover lost Wi-Fi passwords).

🎯 Pro Tips for Hiren's BootCD

✔ Backup data first (use MiniTool Partition Wizard).

✔ Disable Secure Boot in BIOS if tools fail to load.

✔ For UEFI systems, use Hiren's BootCD PE (newer version).

□ *"Which Tool Should I Use First?"*

Problem Type	First Tool to Try
Won't Boot	NTBOOTautofix or bootrec
BSOD	BlueScreenView + MemTest86
Virus	Kaspersky Rescue
Deleted Files	Recuva
Password Reset	NTPWEdit

🚀 Final Advice

- Hiren's BootCD + Ventoy = Ultimate Repair Swiss Army Knife.

- Most fixes take <10 minutes if you know the right tool!